

Managing Multi-Agent Research Systems: A Dashboard for Human Oversight of Coordinating AI Agents

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Try the demo

Motivation

LLMs can generate research ideas rated as novel by human experts (Si et al. 2024), but evaluations stop at the proposal stage—they do not run the experiments. **Execution is where human oversight becomes most critical:** bugs, numerical instabilities, and unexpected results require judgment that idea-level evaluation cannot capture.

We report a **six-week reflective deployment** of a multi-agent research system that generated and executed **200+ proposals and experiments**.

Key Challenges Observed

- Volume:** Agents run 24/7 producing overwhelming information
- Parallelism:** Multiple agents amplify the oversight burden
- Micro vs. macro:** Steering one agent on one task differs from standardizing exchangeable artifacts
- Post-completion review:** Lifting the human-review bottleneck requires dedicated affordances

System Architecture

Agent Roles

T Trick Search Agent

Parses literature, extracts computational primitives (matrix factorizations, parallelization schemes, algebraic structures) as structured "trick cards."

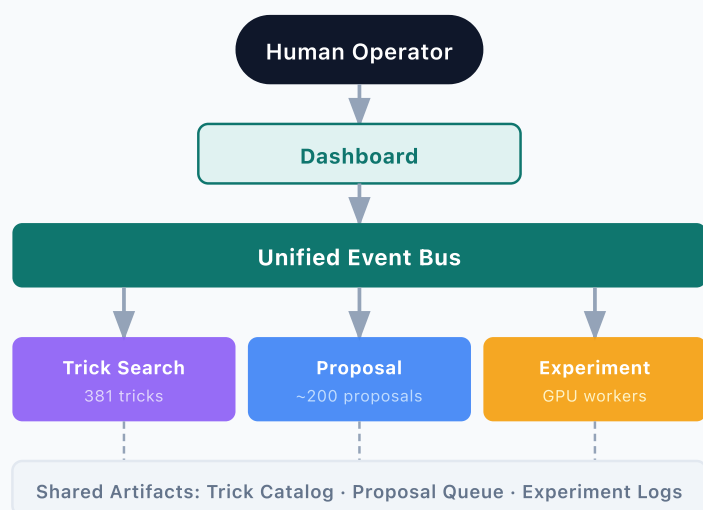
P Proposal Agent

Synthesizes tricks into experiment proposals with hypothesis, formulation, MVE design, and success criteria.

E Experiment Agent

Claims proposals, implements MVEs, runs experiments on GPU infrastructure, reports results. Runs in parallel.

Coordination via Event Bus



Contributions to HEAL Themes

- ✓ **Async-first design:** Multi-agent oversight must assume asynchronous human interaction, with affordances for catching up, triaging, and intervening after the fact.
- ✓ **Multi-level steering:** Micro steering (individual agent/task) vs. macro steering (overall direction) require distinct interface affordances.
- ✓ **Infra vs. behavior monitoring:** Separating "is it running?" from "is it doing the right thing?" was critical for effective debugging.
- ✓ **Human agency through intervention:** What affordances let humans remain active participants alongside autonomous agents?

Figure 1: Dashboard Overview

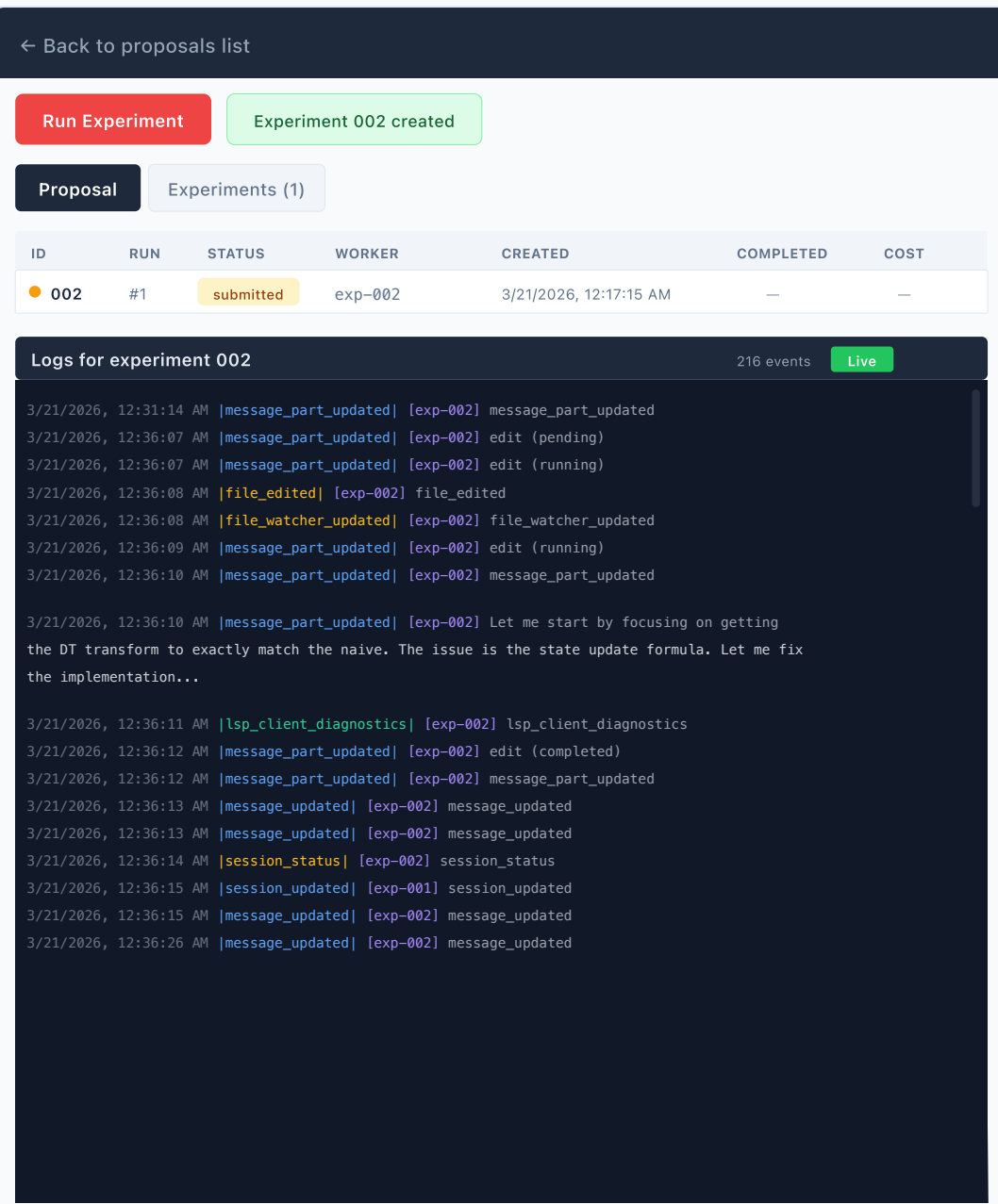


Figure 1: Experiment overview with real-time event stream, color-coded by type.

Figure 2: Experiment Results View

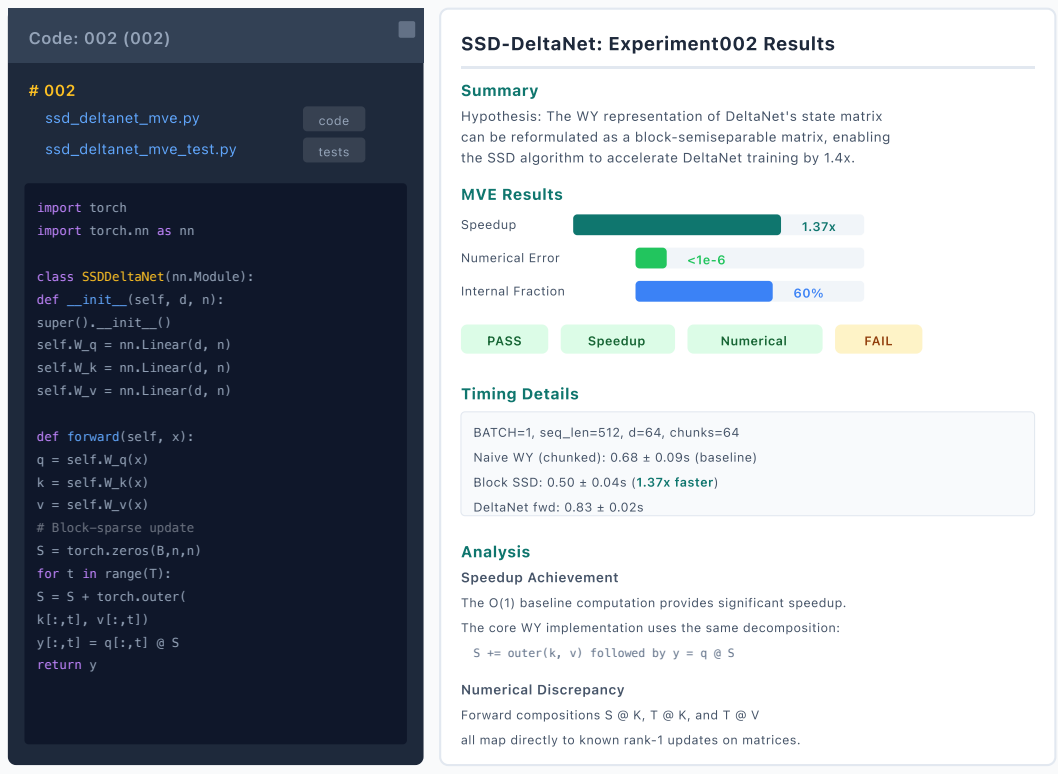


Figure 2: Results view — code, metrics, and artifacts designed for rapid human verification.

Figure 3: Micro-Steering View

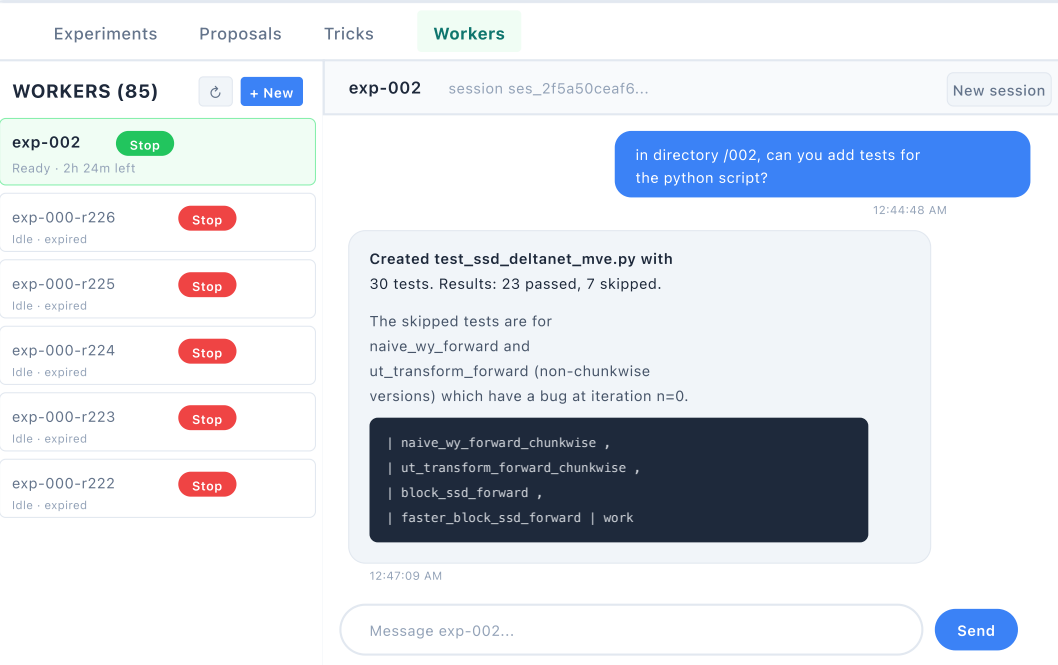


Figure 3: Micro-steering view — an operator can ask an agent for follow-up tasks.

Dashboard Affordances

Asynchronous Review

Timeline surfaces what happened while the user was away: tricks discovered, proposals generated, experiments completed or failed. Filter by time, agent, or event type for rapid triage.

Infrastructure vs. Agent Monitoring

Separating "is it running?" (OOM crashes, network timeouts) from "is it doing the right thing?" (proposal drift, off-topic work) reduced false diagnoses in debugging sessions.

Manual Interventions

Users can kick off experiments, bypass the proposal queue ("queue-jumping"), and add ad-hoc tricks or proposals. Preserves operator agency alongside autonomous operation.

Feedback Throughout Lifecycle

Human input at any point: approving tricks, editing proposals, retrieving artifacts, asking workers for follow-up tasks. Artifacts packaged for inspection of code, traces, and outputs.

Design Implications

IMPLICATION 1

Workflows must support asynchronous interactivity

The event stream quickly becomes overwhelming. Batch updates, async artifact reviews, and alerts for urgent items are essential.

IMPLICATION 2

Use a unified event bus for coordination & context

A shared event stream lets operators spin up new agents for arbitrary tasks—e.g., “summarize progress across all experiments”—without modifying existing infrastructure.

IMPLICATION 3

Separate infrastructure from behavior monitoring

Many "agent errors" were actually OOM crashes or misrouted jobs. Separating these concerns lets operators query system health independently from research progress.

IMPLICATION 4

Design research artifacts to minimize verification time

Agents must produce *reviewable* results—inspectable code, retrievable artifacts, and faithful-to-proposal outputs. Jupyter notebooks proved especially effective, bundling code, results, and analysis into one re-executable artifact.

References

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- [2] Lu et al. "The AI Scientist: Towards Fully Automated Open-ended Scientific Discovery." arXiv:2408.06292, 2024.
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- [5] Amershi et al. "Guidelines for Human-AI Interaction." CHI, 2019.